

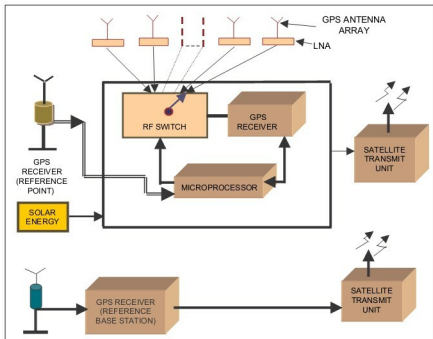


Fig. 3: GPS multi-antenna deformation monitoring transmit system

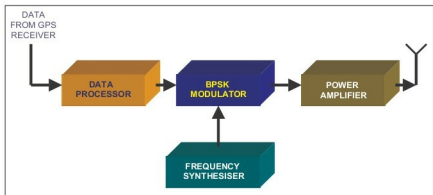


Fig. 4: Satellite transmit unit

on the pseudo-range observable. The accuracy of single-point positioning increases with the number of satellites available.

Relative positioning is the determination of a ground position using two or more receivers and at least four communication satellites. It allows for elimination of clock and atmospheric errors in the carrier-phase signal by combining simultaneous observables (referred to as differencing in the GPS terminology) from multiple receivers and satellites during post-processing. Relative positioning determines the precise vector (baseline) between receiver positions.

When the coordinates of one of the receiver positions are known, that receiver is referred to as a base station, and the known coordinates and baseline can be used to determine the precise coordinates of the unknown points.

In relative positioning, data is stored in the receivers and post-processed using computer software to calculate baselines, determine unknown coordinate positions, and estimate horizontal and vertical errors for calculated positions. If baselines are calculated from two or more base stations, the baselines and coordinates of unknown points can

be further refined through the use of a least-squares adjustment performed by holding the base station positions fixed while adjusting the baselines and coordinate positions of the unknown positions.

A requirement of relative positioning is that the receivers should be capable of recording at least one of the carrier phases. Although many types of receivers are available, dual-frequency receivers are commonly used for relative positioning because these record all GPS codes as well as L1 and L2 carrier phases. Dual-frequency receivers allow ionospheric correction to the GPS signal and improve position determination accuracy.

Landslide monitoring system

GPS has already proven to be an efficient tool for monitoring dam deformations and stability of high-risk slopes. It offers greater accuracy than other surveying techniques. However, GPS has its disadvantages when employed for slope and dam monitoring. One major drawback has been high cost, as large-scale GPS deployments are required in monitoring sites. The conventional GPS monitoring methods, where a permanent GPS receiver must be located at each point, have significant limitations of the cost.

A new approach for deformation monitoring has been developed. It uses a single GPS receiver linked to multiple antennae mounted at different monitoring points. The same is being used for Xiaolangdi dam on the Yellow River in China. A dedicated switching device has been developed for this approach at Hohai University, Nanjing, China. Field-testing results show that the dedicated switching device for GPS multi-antenna system delivers excellent performance and post-processing positioning accuracy is 1-2 mm.

Here I use a similar approach for collecting real-time data from differ-